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Real Estate Agency Portal

Project Purpose

The Real Estate Agency Portal was developed as a web-based platform using PHP and MySQL to support communication and interaction between agents, buyers, and renters. The system allows agents to manage property listings while buyers and renters can browse available properties, save favorites, and submit inquiries about properties they are interested in. The main goal of the project was to create a realistic and user-friendly real estate environment that demonstrates database connectivity, secure login handling, and role-based access control.

Database Design

The project uses a MySQL database named `real_estate_portal_db`. The database was designed using five relational tables: Users, Properties, Inquiries, Transactions, and Favorites. The tables maintained data integrity through primary and foreign key relationships, which established connections between their respective fields. The system used `AddOrUpdateUser` and `ProcessTransaction` stored procedures to facilitate user management and transaction processing. The system created `PropertyListingView` as a view that presents combined listing and agent information in an accessible format. At the same time, the `AfterTransactionInsert` trigger automatically updates property status after a sale or rental transaction.

PHP Functionality

The platform was built using multiple PHP pages, which included `index.php`, `login.php`, `register.php`, `properties.php`, `add_property.php`, and `dashboard.php`. The system has core features

that enable users to create new accounts, add property listings, make inquiries, and access their personal information. The database operations used prepared statements to enhance security while decreasing the threat of SQL injection attacks.

Login Sessions and Role-Based Access

A secure login system was created through the implementation of PHP functions, which include the password_hash function and the password_verify function. The system used session management to track user login time while it protected restricted content. Agents received property management access, while buyers and renters could view listings, store their preferred properties, and make inquiries. The system security improves because sessions automatically terminate when users log out.

Challenges and Solutions

The development team encountered its first obstacle when developers needed to create new user accounts for incoming inquiries without generating duplicate user records. The development team resolved this challenge through the implementation of the AddOrUpdateUser stored procedure. The second challenge required developers to test databases for their ability to maintain correct table relationships through foreign key constraints. The development team used platform screenshots to show they had successfully implemented the system.